PROJECT 40467361

REVISION NOTE

Deployed-Outrigger Verification — Changes from Rev B to Rev C

1 Tonne Mobile Floor Jib Crane | 12 June 2026

Two inputs changed between Rev B and Rev C, and the rest of the report follows from those. The crane material is now Q235, and the deployed lateral track is taken from the updated general-arrangement drawing. A separate editorial change removes the proof-load-test references at the client's request.

1. Material: S275 to Q235

The crane material is now Q235 (Chinese carbon structural steel, GB/T 700), with a nominal yield of 235 MPa, in place of the S275 (275 MPa) assumed in the earlier revisions. This is about a 15 percent drop in yield. All sections are 6 mm wall, well under 16 mm, so the full 235 MPa applies with no thickness de-rating. The member allowable falls from 183.3 MPa to 156.7 MPa. Because the applied stresses do not change, every stress-based factor of safety scales down with the steel.

2. Deployed track: 2376 mm to 2500 mm

The deployed lateral track is now read directly off the updated general-arrangement plan view, which dimensions the foot-to-foot width at 2500 mm. This replaces the 2376 mm that earlier revisions derived from the bare 750 mm leg-arm projection. The half-track is now 1250 mm. The inner frame width is dimensioned at 1100 mm, and the 2360 mm wheelbase is unchanged.

3. Effect on the results

The table below sets the Rev B and Rev C factors side by side. The boom outer and the base outrigger arm, which passed or were marginal on S275, now fail on Q235. The deployed leg still passes.

Check	Rev B (S275)	Rev C (Q235)
Boom outer (100x140x6)	1.48 Marginal	1.26 Fail
Outrigger arm	1.53 Pass	1.30 Fail
Column, combined N+M	0.99 Fail	0.85 Fail
Column-to-base weld	0.85 Fail	0.73 Fail
Outrigger weld	1.29 Fail	1.11 Fail
Deployed leg member	1.91 Pass	1.64 Pass
Pins (Phi30 / Phi20 / hook)	2.44 / 1.20 / 4.03	2.08 / 1.03 / 3.44
Sideways stability, worst	1.05 @ 2376 mm	1.17 @ 2500 mm

What did not move

- Forward stability stays 0.96 (Fail) at the governing mid-reach point. A moment balance does not depend on the steel grade or the lateral track.

- The slewing bolts (1.69) and the hook pin (3.44) are Grade 8.8 fasteners, so they do not move with the base-steel grade.
- The 980 mm front overhang and the 2360 mm wheelbase are unchanged.
- The failing-check count stays at 8 of 14, the layout, Times New Roman and black-and-white styling all carry over.

4. Editorial: proof-load test references removed

At the client's request, all references to the 150 percent proof-load test were removed, since this is a theoretical calculation report. The former reconciliation section is now framed as a purely theoretical elastic-versus-plastic capacity check, with a note that physical load testing is carried out separately with the approval officer.

5. Net conclusion

The two input changes do not alter the overall conclusion: the crane still cannot be signed off to the full 1 tonne chart on a working-stress basis. The weaker Q235 steel makes the structural margin thinner and pulls two further members below target, while the wider 2500 mm track lifts sideways stability slightly but still leaves it short of the 2784 mm needed at mid reach. Forward tipping and the governing column and weld items are unchanged in character. The route forward — re-rating the chart or upgrading the column — is unchanged.